PSI Center for Scientific Computing, Theory and Data

Iodine deposition in lead–bismuth systems

Conservative CFD transport and thermochemical coupling
Project review and validation programme

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Villigen PSI, 7 October 2026 | Working draft

A working solver; physical validation is open

15

Baseline cases passed

60 s; 18,432 cells; 2 ms steps

IMPLEMENTED

Multi-species transport, equilibrium and conservative wall exchange.

5 / 5

Q1 peak temperatures

Within published uncertainty

PROVISIONAL

Full profile agreement and predictive source chemistry remain unresolved.

64

Follow-up cases

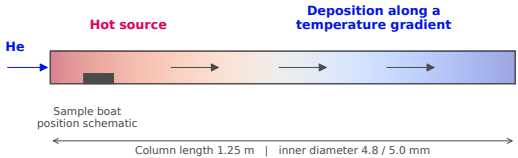
Independent refinement and sensitivities **IN PROGRESS**

16 running; 47 queued. Snapshot: 07 October 2026, 13:55 CEST.

The next decision is whether the deposition profiles are numerically stable and which physical assumptions control them.

Evidence: run/034-liu2025/full_analysis/analysis.json; status_snapshot.json. Q1 uses prescribed iodide release.

Experiment: a source and a temperature gradient



Published operating conditions

He flow: 100 mL/min for pure iodides; 45 mL/min for LBE–I.

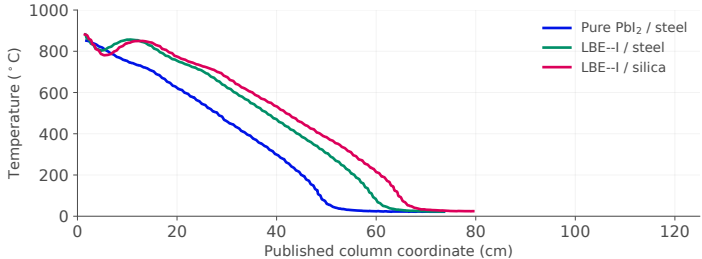
Three-hour exposure, then shutdown and cooling.

Measured response

Axial iodine scan after cooling; deposit characterisation supports PbI_2 and BiI_3 identification.

Liu et al. (2025), DOI: 10.1007/s10967-025-10246-4, experimental section and Tables 2–5. Original schematic; not to scale. Current CFD: empty tube, 4.8 mm steel and 5.0 mm silica.

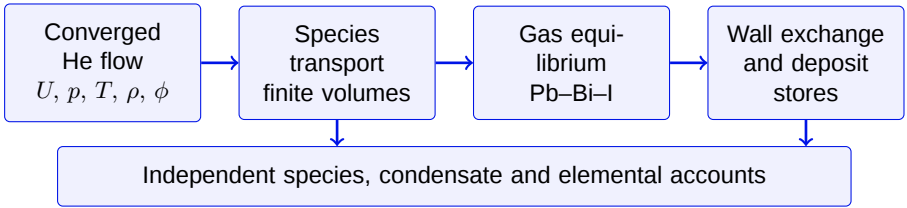
Three published temperature profiles are available



Temperature curves and iodine scans were digitised with their published coordinates; no alignment was fitted.

Liu et al. (2025), published figures; thermochemistry/benchmarks/liu2025/digitized/. BiI3_SiO2_800 lacks a plotted temperature profile and remains unrun.

A frozen carrier supports transient chemistry



- Carrier momentum and temperature stay fixed during species transport.
- Current column matrix uses the gas kernel; GEMS frozen/local modes were qualified separately.
- Formation constants, source rates and solver inputs are fingerprinted.

rhoFixedFlowFoam implementation; phase-change-plan.md, sections 36.9–36.11 and 36.14–36.16. Diagram shows model components, not the exact operator sequence.

Transport and phase change conserve material

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla\cdot(\rho \mathbf{u} Y_i) - \nabla\cdot(\rho D_i \nabla Y_i) = S_i$$

Transport

- Frozen mass flux
- Diffusion and upwind advection
- Accounted source release

Wall exchange

- Hertz–Knudsen–Schrage kinetics
- Half-cell diffusion resistance
- Bounded wall inventories

Every transfer is booked consistently in gas, condensed material and Pb / Bi / I element balances.

applications/rhoFixedFlowFoam/readme.md; solvePairedGasSpecies.H and wall-channel accounting. Carrier He is separate from the nine reacting gases.

Chemistry couples iodides and shared metal stores

Nine transported reacting gases

Pb	PbI	$\mathrm{PbI_2}$
Bi	$\mathrm{Bi_2}$	BiI
$\mathrm{BiI_3}$	I	$\mathrm{I_2}$

Helium supplies the carrier flow. Gas equilibrium preserves local elemental amounts.

Condensed inventories

- $\mathrm{PbI_2}$ and $\mathrm{BiI_3}$ deposits
- Shared Bi / $\mathrm{Bi_2}$ store
- Pb metal store

Reactive wall channel

$$3\,\mathrm{BiI(g)} \longrightarrow 2\,\mathrm{Bi(s)} + \mathrm{BiI_3(g)}$$

The deposited iodide split depends on source composition, gas speciation and surface reactions.

M10b gas kernel; M10d shared condensates and reactive wall channels. The displayed wall reaction is the implemented irreversible channel.

Two source models answer different questions

Q1: prescribed iodides

PbI_2 and BiI_3 mean release rates are derived from published deposited amounts.

Purpose: assess transport with a source constrained by the measured split.

Q2: atomic release

I, Pb and Bi are supplied; gas chemistry forms iodides. Metal supply is a prescribed saturation fraction.

Purpose: test source-conditioned predictions of the split.

The 60 s CFD window retains the three-hour mean release rate: it supplies 1/180 of the assumed three-hour inventory.

run/034-liu2025 case manifests; Q1/Q2 source definitions in cases.py. The 5-degree wedge contains 1/72 of the full-column supply.

Verification covers equilibrium and conservation

References

- Wall closed forms
- Equilibrium oracle
- Plug-flow onsets

Robustness

- Strong binding
- 600,000 stress inputs
- Guarded fallback

Consistency

- Serial / four ranks
- Binary restarts
- Element ledgers

Follow-up launch checks

64 / 64 native ten-step smoke checks passed; 9 carrier audits passed; 19 Python checks passed.

phase-change-plan.md, sections 36.9–36.16; run/034-liu2025/followup_launch.json. These are targeted checks; no new aggregate Alltest or v2606 qualification is claimed.

All 15 available baseline cases finished

Experiment	Cases	Source / thermodynamics
Pure PbI_2 / steel	1	Q1 / baseline
LBE–I / steel	7	Q1 + six Q2 variants
LBE–I / silica	7	Q1 + six Q2 variants

8.63×10^{-16}
Maximum accounted element closure

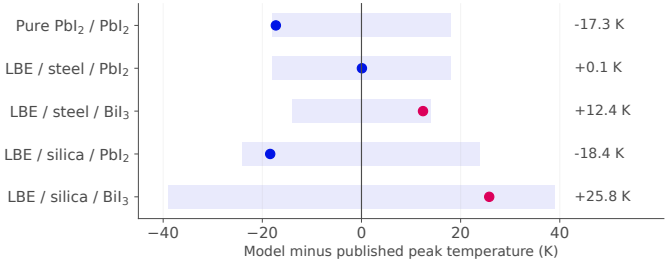
5.98×10^{-13}
Maximum raw material residue / supply

0
Gas-equilibrium solver failures

18,432 cells | 60 s | 2 ms | 15 PASS / 0 FAIL / 1 missing-profile NOTRUN

full_analysis/analysis.json and independent material audits. BiI3_SiO2_800 is the unavailable case. Numerical PASS does not establish experimental validation.

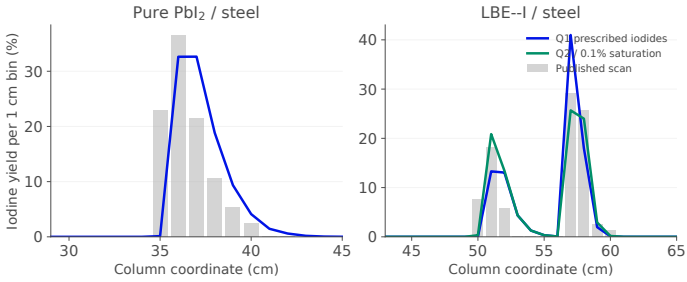
Five Q1 peak temperatures match uncertainty bands



Temperature agreement is encouraging; spatial profiles and source-conditioned species splits require separate assessment.

Liu et al. (2025), published deposition temperatures and uncertainties; full_analysis/q1_peaks.csv. Shaded bands show the published uncertainty around zero error.

Steel profiles reveal shape and split differences

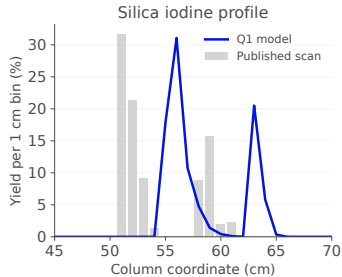


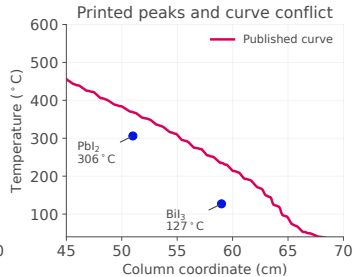
Steel Q2 at 0.1% saturation: BiI_3 iodide-iodine share **56.47%**, compared with **65.63%** in the paper.

The lowest tested steel profile error is still 36.80 percentage points in summed binwise absolute difference.

Digitised published scan heights retained; full_analysis/case_metrics.csv and saved iodine-comparison.csv. The LBE experimental scan includes KI; the model omits K chemistry.

Silica deposition positions differ by 4–5 cm



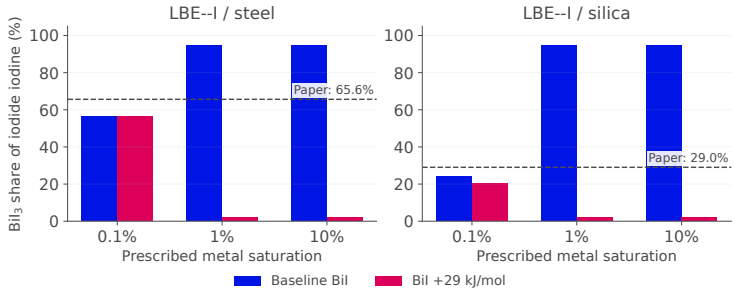


Q1 model / scan overlap is only about **7%** after secondary normalisation over the common scan.

The printed peak temperatures conflict with the published curve at the scan maxima.

Liu et al. (2025), Figure 7 and printed peak temperatures; full_analysis/silica_figure_diagnostics and case metrics. No coordinate correction applied in these plots.

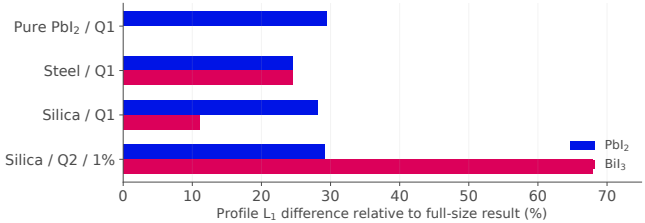
BiI data and source loading can reverse the split



At 1% metal saturation, the baseline and shifted BiI scenarios give roughly 94.5% and 2% BiI_3 shares.

full_analysis/thermodynamic_sensitivity.csv. Share = iodine in BiI3 / iodine in (PbI2 + BiI3). The +29 kJ/mol BiI shift is a surrogate sensitivity, not a second measured thermodynamic record.

Earlier refinement changed several factors together



Previous comparison: 2,048 → 18,432 cells; 20 → 2 ms; solver revision also changed.

Profile changes reach 67.95% L_1. First-order upwind numerical diffusion is comparable to the prescribed physical diffusion.

full_analysis/combined_refinement.csv; transport-scale estimates in analysis.json. This mixed comparison does not establish independent mesh or time convergence.

Independent spatial and temporal refinement is running

Fixed 2 ms time step: three meshes

8,192	18,432	41,472
Coarse	**Baseline**	**Fine**
256 × 32 cells	384 × 48 cells	576 × 72 cells

Fixed 18,432-cell mesh: 2 ms versus 1 ms

Four cases: pure PbI_2, low/high metal loading and shifted BiI data.

Outputs used to decide adequacy

Profile L_1, centroid, amount and split; conditional three-grid estimates.

64-case study manifest; twelve new numerical cases. Uniform grid ratio 1.5 in each direction. Existing full-size results supply the medium grid / 2 ms reference.

The 64-case study exposes provisional assumptions

Study	N	Variation
Independent refinement	12	Mesh / time step
Species diffusion	12	$0.5D$, $2D$, PbI_2–He
Iodide accommodation	8	$\alpha = 0.1,\ 0.01$
Source position	8	± 15 mm
Source loading	12	Loading bracket
Flow reference	4	$0\,^{\circ}\mathrm{C}$ alternative
Silica alignment	4	Two hypotheses
Isothermal flow stop	4	Restart 60–80 s

Frozen execution snapshot

16 running
47 queued
1 full cases passed
0 failed

07 October 2026, 13:55 CEST

Up to 16 concurrent serial jobs; report generated after completion.

followup_study.py; followup_launch.json; dated status snapshot. Accommodation applies to iodide pairs; the empty-tube carrier still omits boat obstruction. No experimental parameter was fitted.

Exposure duration and cooling need separate evidence

Analytical duration

Continue 50–60 s wall rates to 10,800 s.

Assumes steady supply and deposition rates.

Conditional gas iodine inventory:
0.010–0.0485% of three-hour supply.

Isothermal flow-stop CFD

Restart at 60 s; stop source and carrier flow; advance to 80 s.

Closed inventory; fixed temperature.

Tests redistribution after flow stops.

Experimental cooling could redistribute existing wall deposits. The paper supplies no measured temperature-versus-time history.

physical_input_audit/duration_projection.csv (44 conditional profiles); shutdown case manifests; Liu experimental protocol. Neither study reconstructs the actual three-hour exposure plus cooling.

Decisions after the refinement study

Evidence	Decision	Remaining work
Mesh / time profiles	Select an adequate discretisation	Further refinement if needed
Input sensitivities	Identify robust and source-sensitive predictions	Qualify thermodynamic / surface data
Silica hypotheses	Assess effect of coordinate discrepancy	Confirm experimental temperature input
Flow-stop / duration	Bound stated idealised effects	Obtain cooling and release evidence
Completed scientific analysis	Define defensible manuscript claims	Final regression, archive, co-author review

The supplied paper supports a reproducible modelling and sensitivity study; stronger predictive claims require better-constrained inputs.

Physical input audit and M10e scientific acceptance criteria. Missing BiI3_SiO2_800 profile remains explicit. Broader M6/M8 comparison and validation requirements require separate reconciliation.

Conservative model; validation remains open

1. **Implemented:** conservative transport, equilibrium and wall chemistry.
2. **Verified numerically:** 15 baseline runs with tight element balances.
3. **In progress:** profile convergence and physical-input sensitivity.

Next review

Review the 64-case results, select the numerical setup and define supported physical claims.

Primary references
Liu et al., *J. Radioanal. Nucl. Chem.* 334 (2025), 5201–5216. doi:10.1007/s10967-025-10246-4
Lobresco et al., *Nucl. Eng. Des.* 459 (2026), 115201. doi:10.1016/j.nucengdes.2026.115201

Project evidence: baseline analysis, physical-input audit and follow-up launch record, JanKren/LESTO, study commit `361f02d`.